

# Quiz 5-A

## Solution

Date

Q1 Here,  $m = -8$  and  $n = 2$

$$m^2 - n^2 = (-8)^2 - (2)^2 \\ = 64 - 4 = 60$$

and  $5 \nmid 60$ , hence  $(-8)R2$ . ✓

Q2 → Reflexive:  $0 - 0 = 0$  which is even, hence, it is NOT reflexive.  
i.e.  $m \not Q m$  in general.

→ Symmetry: Assume  $mOn$  is true, i.e.  $m - n$  is odd.

$$m - n = 2k + 1 \quad ; \text{ for some integer } k$$

$$\Rightarrow n - m = -2k - 1$$

$$n - m = -2k - 2 + 1$$

$$n - m = -2(k + 1) + 1$$

$$\text{or } n - m = 2(-k - 1) + 1$$

Here,  $-k - 1$  is an integer,

hence  $n - m$  is an odd integer by definition i.e.

$nOm$ . It is symmetric.

→ Transitivity:  $2 - 1 = 1$  is odd hence  $2O1$ ,  $1 - 0 = 1$  is odd hence  $1O0$ , but  $2 - 0 = 0$  is even, hence  $2 \not Q 0$ . Hence, it is not transitive. i.e. in general. If  $mOn$  and  $nOp$ , then it does not imply that  $mOp$ .

Q3 (a)  $R_2$  is transitive but not symmetric. It is transitive since  $(1, 2)$  and  $(2, 3)$  are pairs and so is  $(1, 3)$ . It is not symmetric as it has  $(1, 2)$  but not  $(2, 1)$ . (among other counterexamples).

(b)  $R_3$  is symmetric as it contains  $(1, 2)$  and  $(2, 1)$ ,  $(2, 3)$  and  $(3, 2)$  but not transitive as  $(1, 2)$  and  $(2, 3)$  but there is no  $(1, 3)$ .

(c) (i) It is ~~not~~ reflexive as there ~~are~~ are  $(1, 1)$ ,  $(2, 2)$  and  $(3, 3)$  pairs.

- ii) It is symmetric (as shown in (b))
  - iii) It is not transitive (as shown in (b))
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Hence, it doesn't have all 3 properties.

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